

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 6. Loops - Control Flow

Lecture: 58. Else suite - While

Section 1: Lecture Summary

This lecture covers the Python feature called else suite used with loops, focusing mainly on the while loop. Unlike the commonly known use of else with if statements, Python allows an else clause to be appended to loops (while and for), which is executed only when the loop terminates naturally — that is, when its condition becomes false, and not interrupted by a break statement. The instructor recapitalizes how if-else works and then explains the analogous use of else in while loops with examples. The key insight is that the else block in a loop serves as a signal that the loop has completed all iterations successfully without any premature exit caused by break. Code demonstrations show the difference in output and behavior when a break is introduced, proving when the else block runs or is skipped. This feature is unique to Python and helps write more expressive loop controls.

Section 2: Key Concepts and Explanations

- Else with if statement (Recap)

- Executes if the if condition is false.
- Mutual exclusivity: either if block or else block runs.

- Else with While loop

- Syntax structure:

loop body statements

else block statements

```
while <condition>:  
    else:
```

- The while loop executes repeatedly as long as the condition is true.
- When the condition becomes false, the else block executes immediately after exiting the loop.
- If a break statement interrupts the loop, the else block does NOT run.
- Statements following the loop (outside) always execute after the loop ends, regardless of break.

- Break Statement Effect

- Used to exit the loop prematurely within some inner condition.
- Causes loop termination without the loop condition becoming false.
- Prevents execution of the else block attached to the loop.

- Practical Purpose of Else in Loops

- Indicates that the loop ended successfully (condition false, no break).
- Provides a place to execute code only if the loop completed all iterations.
- Useful for distinguishing between a normal termination vs. early exit.

- Comparison with Other Languages

- This else on loops is a Python-specific feature not present in many other programming languages.
- It makes programs simpler and more expressive.

Section 3: Example Code and Use Cases

```
i = 0
while i < 5:
    i = i + 1
    print(i)
else:
    print('end of loop')

print('end of program')
```

- This script prints numbers 1 to 5, then prints "end of loop" because the loop ended naturally (condition became false), then "end of program" outside the loop.

```
i = 0
while i < 5:
    i = i + 1
    if i == 3:
        break
    print(i)
else:
    print('end of loop')

print('end of program')
```

- Here, when `i` becomes 3, `break` exits the loop immediately.
- The `else` block is not executed, so "end of loop" does not print.
- The output prints "1", "2", then "end of program".

Section 4: Key Takeaways

- Python's else can be used with loops, not just with if statements.

- The else block after a loop runs **only if the loop completes normally** — i.e., the loop condition becomes false without interruption.
- If the loop terminates due to a break, the else block is skipped.
- Statements after the loop (outside the loop) are always executed after the loop ends.
- Using else with loops adds semantic clarity about the termination status of the loop.
- This feature is unique to Python and can make code simpler and clearer in certain scenarios.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



else suite
with While Loop

else suite
with While Loop

- if

else suite

with While Loop

- if
- while

else suite

with While Loop

- if
- while
- for

else suite

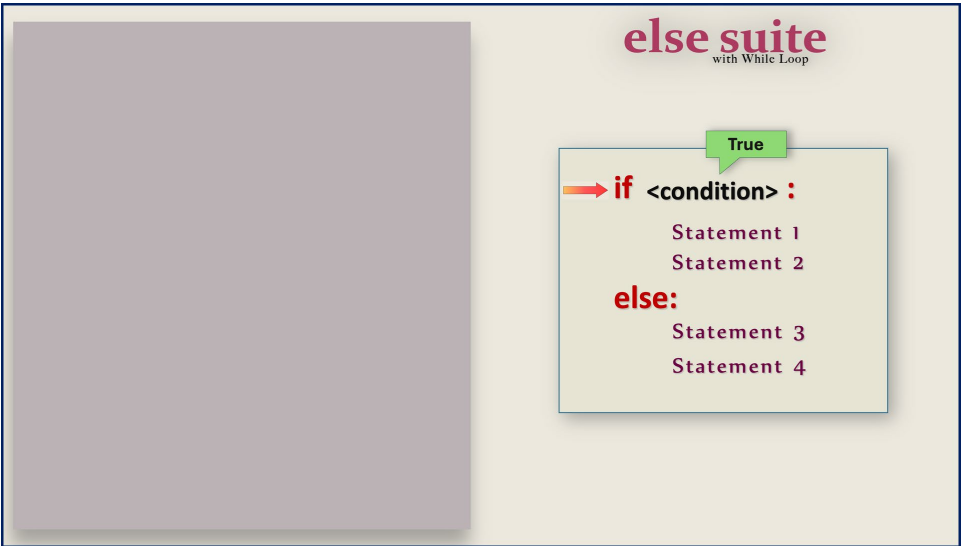
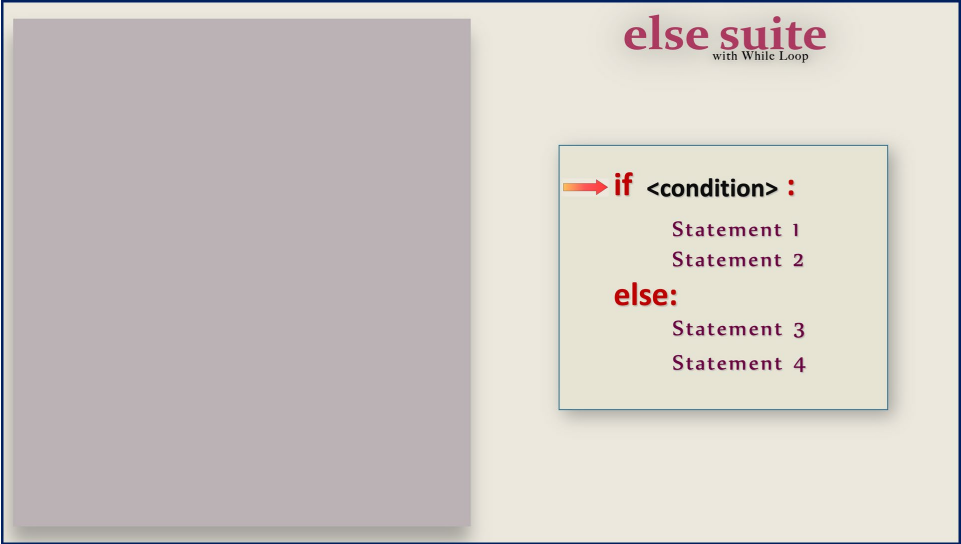
with While Loop

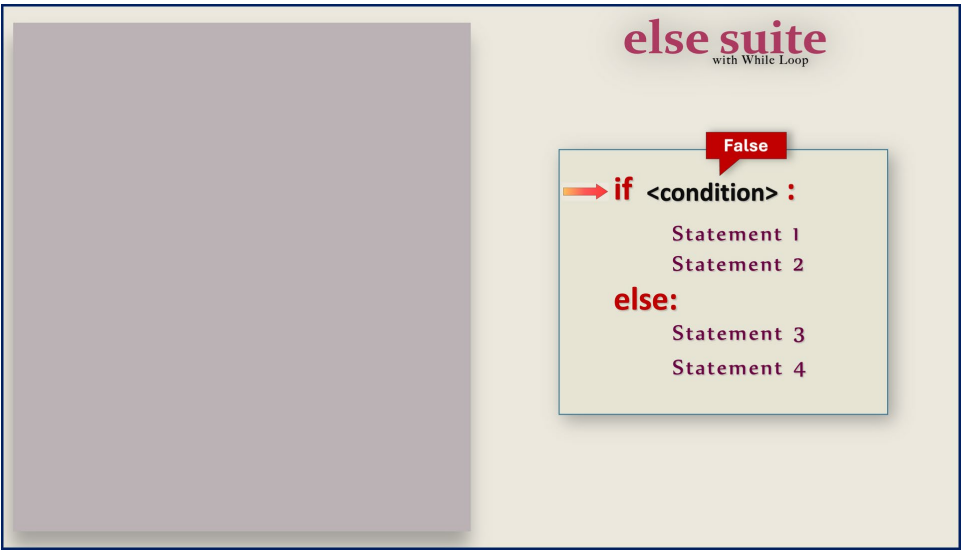
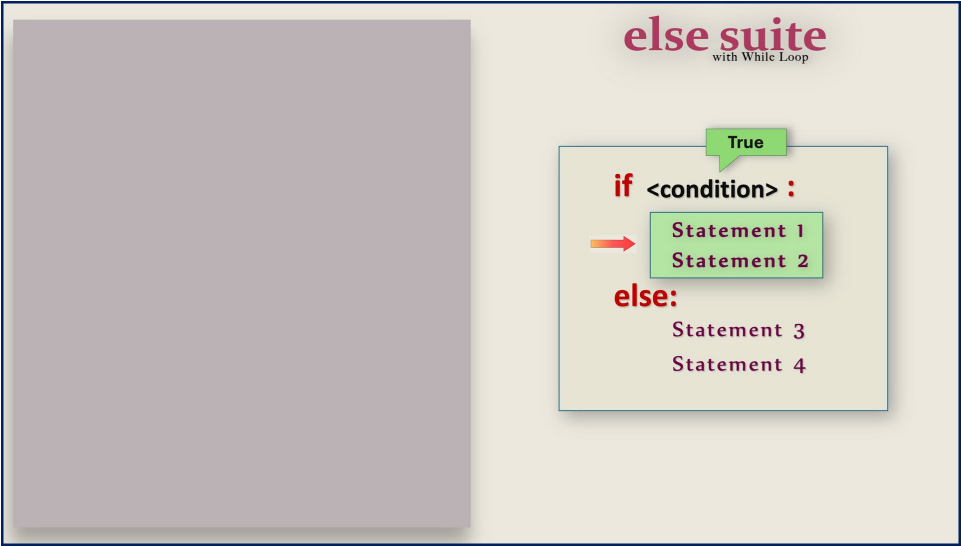
- if
- while
- for
- try

else suite

with While Loop

```
if <condition> :  
    Statement 1  
    Statement 2  
else:  
    Statement 3  
    Statement 4
```



else suite

with While Loop

False

if <condition> :

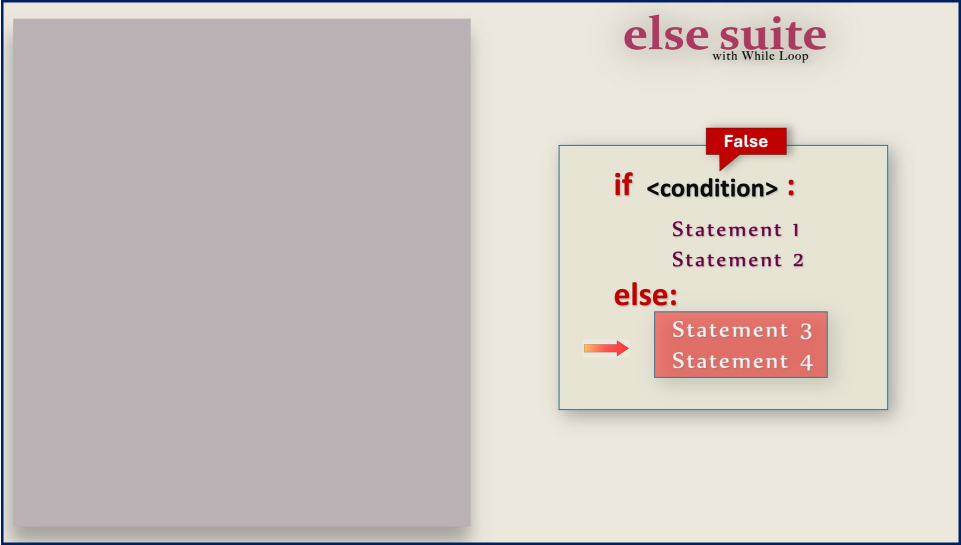
Statement 1

Statement 2

→ **else:**

Statement 3

Statement 4



else suite

with While Loop

while <condition> :

Statement 1

Statement 2

else:

Statement 3

else suite

with While Loop

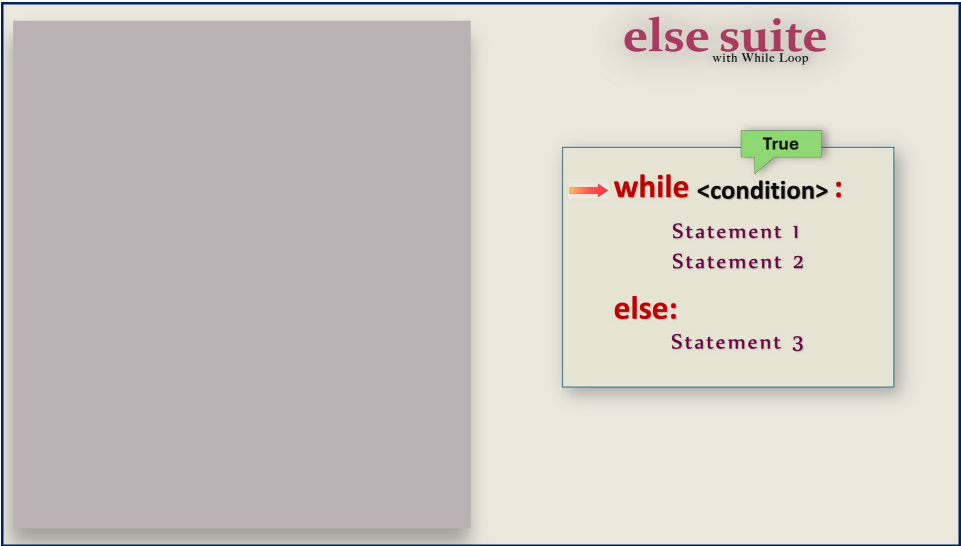
→ **while** <condition> :

Statement 1

Statement 2

else:

Statement 3



else suite

with While Loop

False

→ **while** <condition> :
Statement 1
Statement 2

else:
Statement 3

else suite

with While Loop

False

while <condition> :
Statement 1
Statement 2

→ **else:**
Statement 3



else suite
with While Loop

→ **while** <condition> :

Statement 1

Statement 2

Statement 3



else suite

with While Loop

while <condition> :

Statement 1

Statement 2

else:

Statement 3

else suite

with While Loop

while <condition> :

Statement 1

Statement 2

else:

Statement 3

Statement 4



else suite

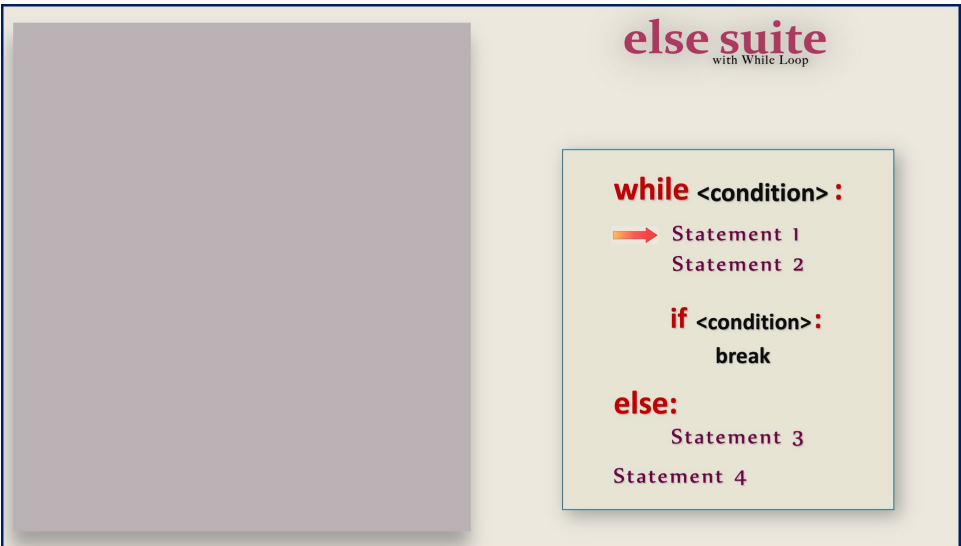
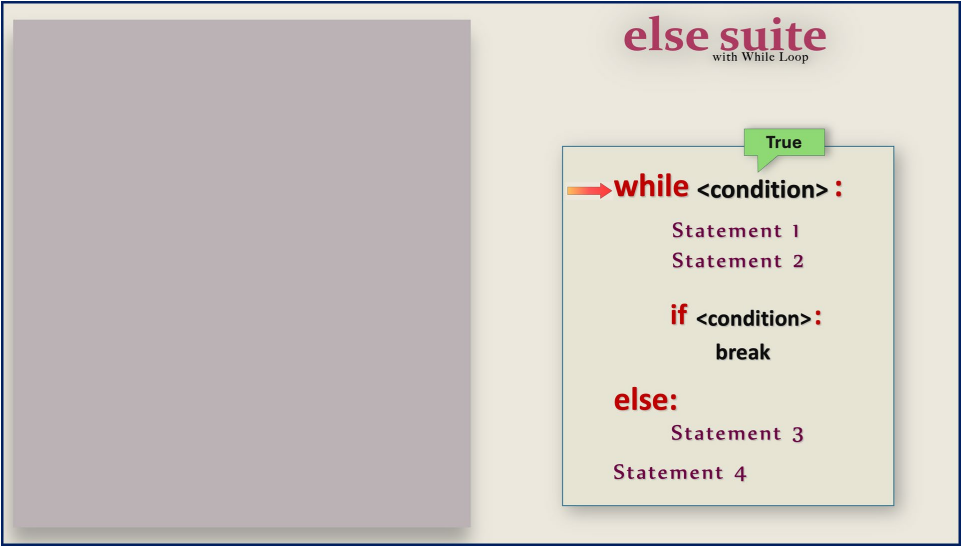
with While Loop

```
while <condition> :  
    Statement 1  
    Statement 2  
  
    if <condition> :  
        break  
  
    else:  
        Statement 3  
Statement 4
```

else suite

with While Loop

```
→ while <condition> :  
    Statement 1  
    Statement 2  
  
    if <condition> :  
        break  
  
    else:  
        Statement 3  
Statement 4
```



else suite

with While Loop

while <condition> :

Statement 1

→ Statement 2

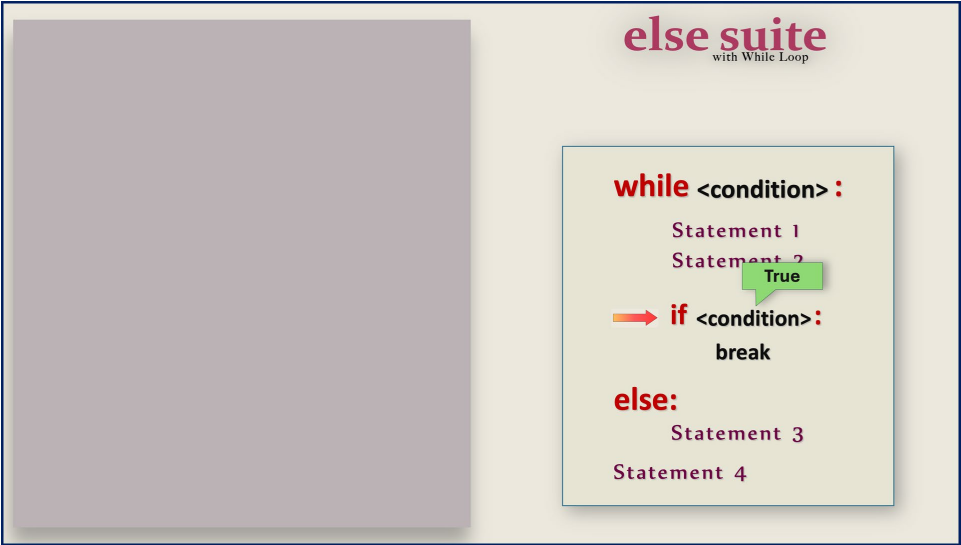
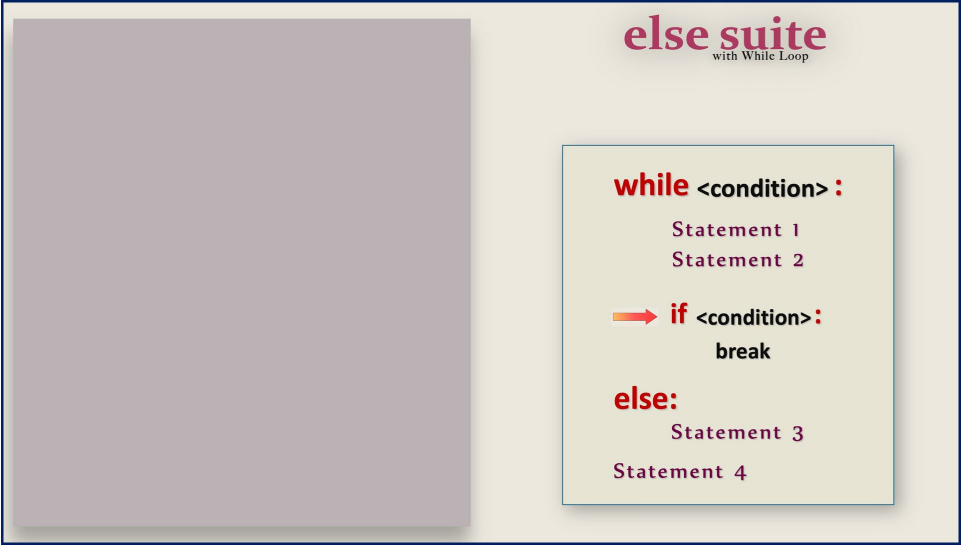
if <condition> :

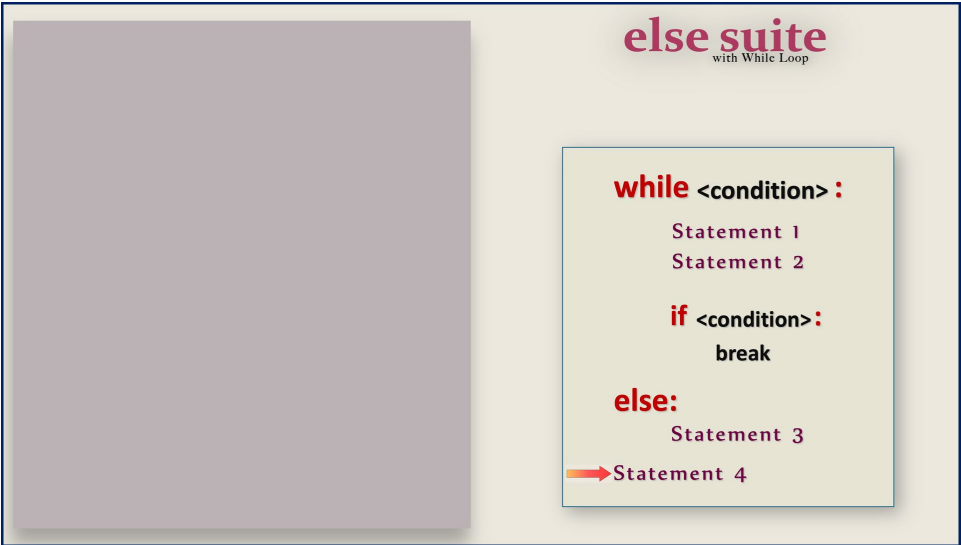
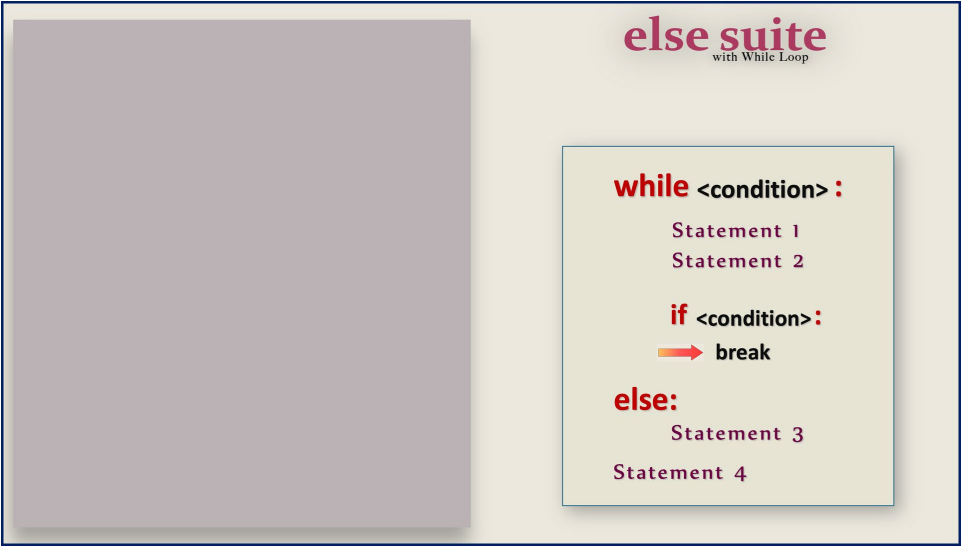
break

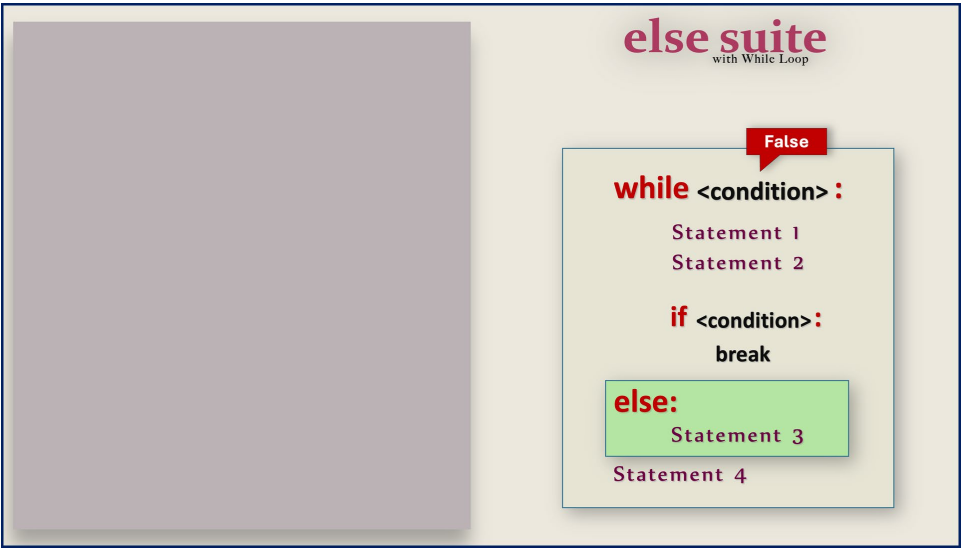
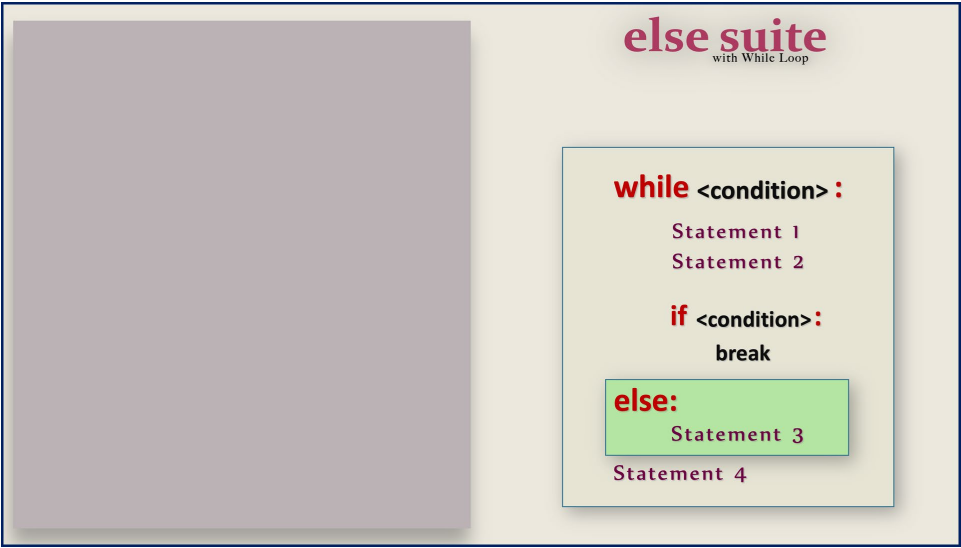
else:

Statement 3

Statement 4







else suite

with While Loop

```
i = 0
while i < 5 :
    i = i + 1
    print(i)

print( 'End of Program' )
```

else suite

with While Loop

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while i < 5 :
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else suite

with While Loop

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else suite

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else suite

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print( 'End of Program' )
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